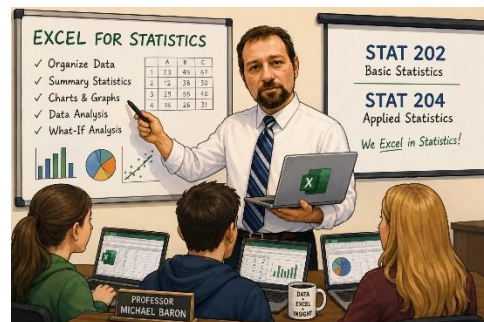


Tests and Confidence Intervals

Critical values



For confidence intervals and testing hypotheses with acceptance/rejection regions, we need to calculate the critical values $Z_{\alpha/2}$ and $t_{\alpha/2}$ (Z_{α} and t_{α} for one-sided tests).

Use commands

`=NORMINV( 1- $\alpha$ , 0, 1 )` to get Z_{α} for the upper-tail probability α

`=NORMINV( 1- $\alpha/2$ , 0, 1 )` to get $Z_{\alpha/2}$ for the upper-tail probability $\alpha/2$

`=TINV(  $\alpha$ , df )` to get $t_{\alpha/2}$ for the upper-tail probability $\alpha/2$

(This command gives a two-sided critical value; it will automatically divide α by 2)

`=T.INV(  $\alpha$ , df )` or `=TINV( 2* $\alpha$ , df )` to get t_{α} for the upper-tail probability α

Use the correct number of degrees of freedom df (n-1 for a one-sample test, n1+n2-2 for a two-sample test with equal variances, etc.).

Pvalues

P-value is always the probability to see a test statistic *at least as extreme* as the one that we have really observed. By “extreme” we mean contradicting the null hypothesis H_0 in favor of the alternative H_1 . It is a probability, and to find it, we use the appropriate distribution of the test statistic.

For a **Z-test** with a test statistic Z_{stat} , use the Standard Normal distribution

`=NORMDIST( $Z_{\text{stat}}$ , 0, 1, 1)` to get the p-value for the left-tail alternative $H_1: \mu < \mu_0$

`=1-NORMDIST( $Z_{\text{stat}}$ , 0, 1, 1)` to get the p-value for the right-tail alternative $H_1: \mu > \mu_0$

`=2*NORMDIST(-abs( $Z_{\text{stat}}$ ), 0, 1, 1)` to get the p-value for the two-sided alternative $H_1: \mu \neq \mu_0$

To get the right-tail p-value, we used the complement, because the NORMDIST command always gives the left-tail probability. For the two-sided test, just take the smaller tail probability and double it. If Z_{stat} is negative, you can use `=2*NORMDIST( $Z_{\text{stat}}$ , 0, 1, 1)`. If Z_{stat} is positive, use `=2*NORMDIST(- $Z_{\text{stat}}$ , 0, 1, 1)`. I wrote the absolute value “abs” to cover both cases in one command.

For a **t-test** with a test statistic t_{stat} , use the Student's t-distribution with the correct degrees of freedom. The command is **=TDIST(t_{stat} , df, tails)**, where **tails=1** for a one-sided test and **tails=2** for a two-sided test. This command can only handle $t_{\text{stat}} > 0$, and it returns the upper-tail probability. Therefore:

For the *right-tail alternative* $H_1: \mu > \mu_0$, use

=TDIST(t_{stat} , df, 1) to get the p-value, when $t_{\text{stat}} > 0$ (in this case, p-value < 0.5)

=1-TDIST(- t_{stat} , df, 1) to get the p-value, when $t_{\text{stat}} < 0$ (p-value > 0.5, H_0 is accepted)

For the *left-tail alternative* $H_1: \mu < \mu_0$, use

=1-TDIST(t_{stat} , df, 1) to get the p-value, when $t_{\text{stat}} > 0$ (p-value > 0.5, H_0 is accepted)

=TDIST(- t_{stat} , df, 1) to get the p-value, when $t_{\text{stat}} < 0$ (p-value < 0.5)

For the *two-sided alternative* $H_1: \mu \neq \mu_0$, use

=TDIST(t_{stat} , df, 2) to get the p-value, when $t_{\text{stat}} > 0$

=TDIST(- t_{stat} , df, 2) to get the p-value, when $t_{\text{stat}} < 0$

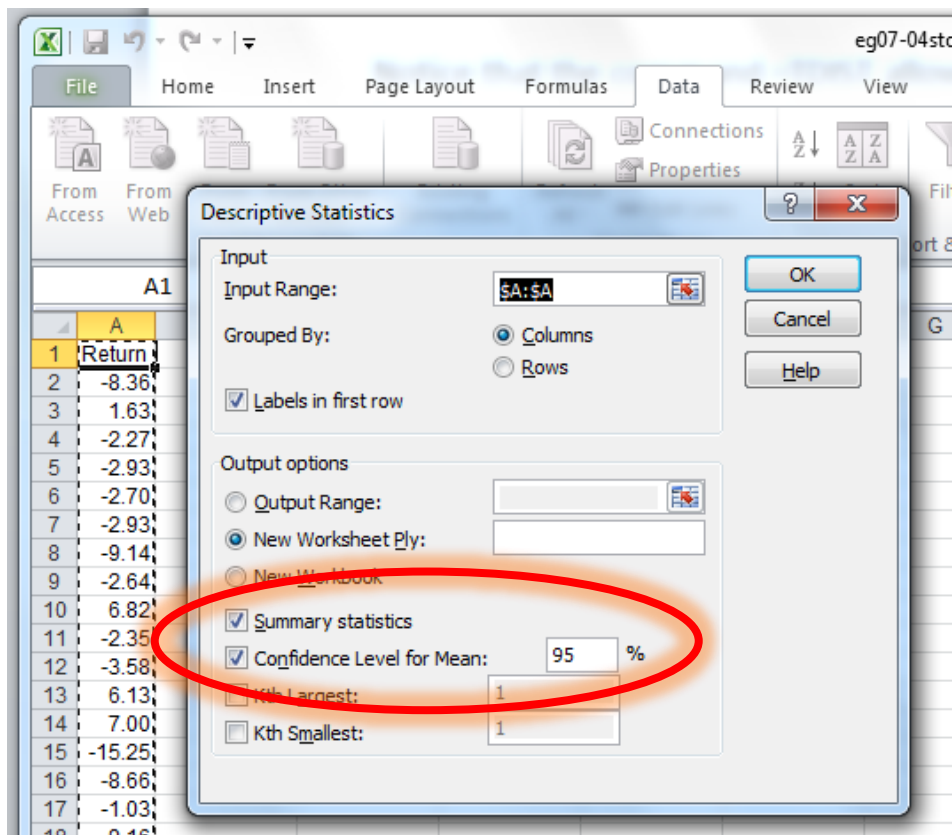
To summarize, the command **=TDIST** allows only positive values of t_{stat} . If your t_{stat} is negative, use **(- t_{stat})** instead. Also, notice that the p-value > 0.5 when the left-tail test $H_1: \mu < \mu_0$ has $t_{\text{stat}} > 0$, and when the right-tail test $H_1: \mu > \mu_0$ has $t_{\text{stat}} < 0$. In both of these cases, there is absolutely no support of the alternative, so the p-value is high, and the null hypothesis H_0 is not rejected.

The Data: sample mean, standard deviation, etc.

To get the sample statistics required for tests and confidence intervals, you can refer to one of our past Excel handouts - *Descriptive Statistics*. Here are the relevant Excel commands

Statistic	Excel function
Sample mean $\bar{x}$	=AVERAGE(A2:A55)
Sample standard deviation s	=STDEV(A2:A55)
Sample size n	=COUNT(A2:A55)

You can get all of these (and more!) from the *Descriptive Statistics* tool of the Data Analysis Toolpak:



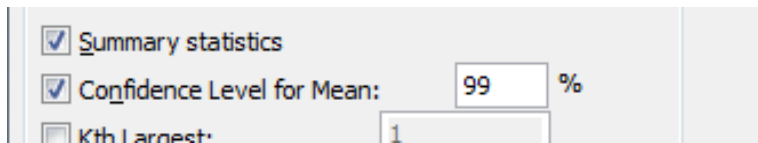
Notice the new box "Confidence level" that I checked...

	A	B	C
1	Return		
2			
3	Mean	-1.1	
4	Standard Error	0.959	
5	Median	-1.41	
6	Mode	-2.93	
7	Standard Deviation	5.991	
8	Sample Variance	35.89	
9	Kurtosis	0.227	
10	Skewness	0.159	
11	Range	27.47	
12	Minimum	-15.25	
13	Maximum	12.22	
14	Sum	-42.89	
15	Count	39	
16	Confidence Level(95.0%)	1.942	
17			
18			

As a result, we get the sample mean $\bar{x} = -1.1$, the sample standard deviation $s = 5.991$, and the sample size $n = 39$. Also, there are some shortcuts:

Standard Error 0.959 is in fact $Std(\bar{x}) = s / \sqrt{n}$

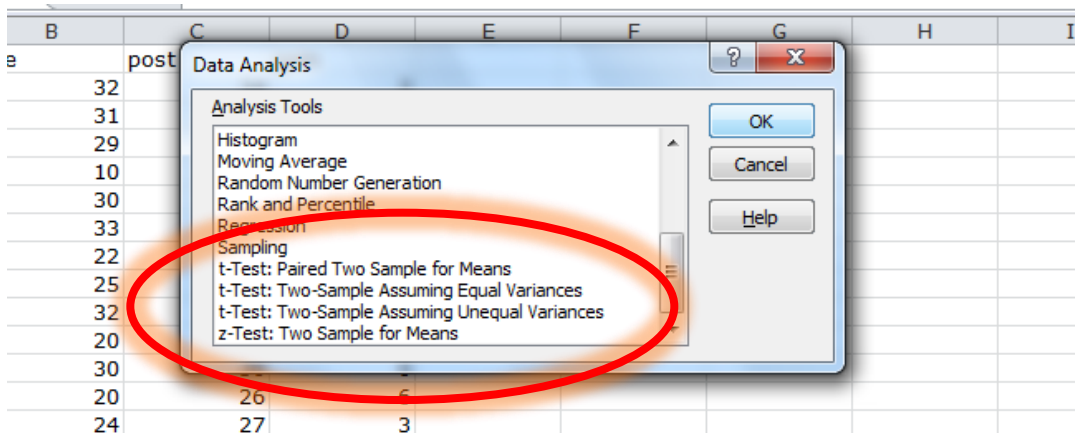
Confidence Level (95.0%) is $(t_{\alpha/2} s / \sqrt{n})$, the margin of the 95% confidence interval for μ . Need a 99% or a 88% confidence interval? Enter that confidence level in the descriptive statistics toolbox instead of 95%.



Two-sample procedures in the Data Analysis Toolpak

All of the above can be applied to two-sample tests and confidence intervals for $(\mu_1 - \mu_2)$.

Also, the Data Analysis Toolpak has procedures for two-sample Z-tests, t-tests, and F-tests. Separate tools are for the case of equal variances and unequal variances.



These tests will give you an output similar to this.

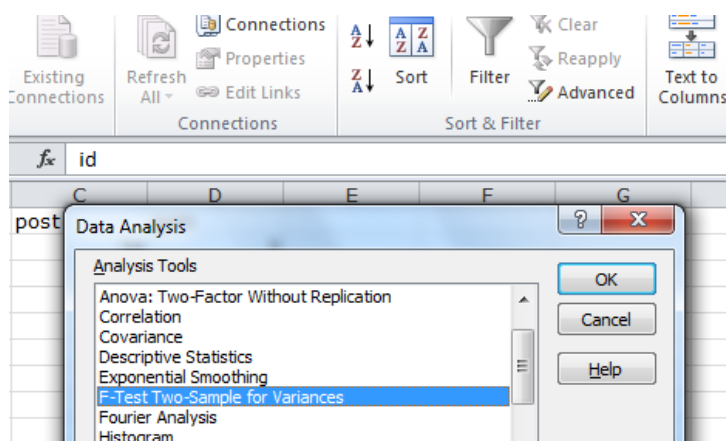
	A	B	C
1	t-Test: Two-Sample Assuming Unequal Variances		
2			
3		Variable 1	Variable 2
4	Mean	7.69655	7.28164
5	Variance	2.9612	5.37772
6	Observations	31	47
7	Hypothesized Mean Difference	0	
8	df	75	
9	t Stat	0.90553	
10	P(T<=t) one-tail	0.18404	
11	t Critical one-tail	1.66543	
12	P(T<=t) two-tail	0.36808	
13	t Critical two-tail	1.9921	

P-values are denoted by $P(T \leq t)$. As we know, the two-sided P-value is simply twice the one-sided P-value. Also, there are critical values from the t-table, $t_\alpha = 1.66543$ and $t_{\alpha/2} = 1.99210$, computed for the level α specified as “alpha” in the toolbox.

This is the *unequal-variance* t-test, and its df are computed by the Satterthwaite approximation. For the *equal-variance* test, $df = n_1 + n_2 - 2$, and also, the pooled variance will be computed.

F-test comparing two variances

We can use the F-test to decide if the assumption of equal variances is reasonable. It is used to test $H_0: \sigma_1 = \sigma_2$ vs $H_1: \sigma_1 \neq \sigma_2$



Result of the F-test looks like this.

	A	B	C
1	F-Test Two-Sample for Variances		
2			
3		<i>Variable 1</i>	<i>Variable 2</i>
4	Mean	7.696548	7.281638
5	Variance	2.961198	5.377723
6	Observations	31	47
7	df	30	46
8	F	0.550642	
9	P(F<=f) one-tail	0.043217	
10	F Critical one-tail	0.564408	
11			

Here, the test statistic $F=0.564408$ and the one-sided Pvalue = $P(F \leq 0.564408) = 0.043217$. You can also obtain this P-value by the command **=1-Fdist(0.564408,30,46)**. Notice, however, that for testing $H_0: \sigma_1 = \sigma_2$ vs $H_1: \sigma_1 \neq \sigma_2$, a two-sided test, the P-value is $2*0.043217 = 0.086434$.